

AI ETHICS & REGULATIONS

INSTRUCTOR:

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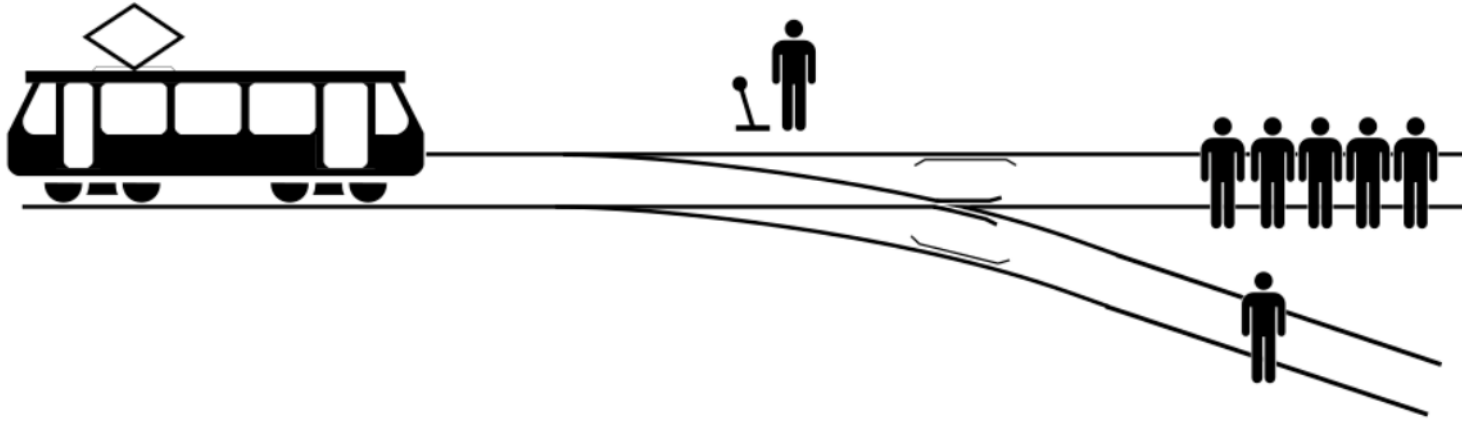


MORAL THEORIES:

APPLICATIONS IN ETHICS LIMITATIONS

- Introduction
- Review of ethical theories
- Comparing workable ethical theories

INTRODUCTION



The trolley problem presents a dilemma: is it preferable to pull the lever to divert the runaway trolley onto the side track with just one person?

Source: Wikipedia

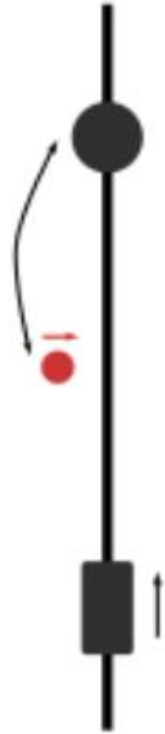
THE TROLLEY PROBLEM



the switch
Foot, 1967



the fat man
Thomson, 1976



the fat villain

THE TROLLEY PROBLEM (VARIATIONS)

Source: Wikipedia

THE ETHICAL POINT OF VIEW



Human lives in communities / societies



Most everyone shares
“core values”,
desiring:

Life
Happiness
Ability to accomplish goals

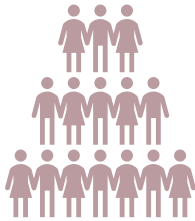


Two ways to view
world

Selfish point of view:
consider only your own self
and your core values

Ethical point of view:
respect other people and
their core values

COMMON DEFINITIONS



Community / Society

Association of people organized under a system of rules

Rules: advance the good of members (as a whole) over time



Morality

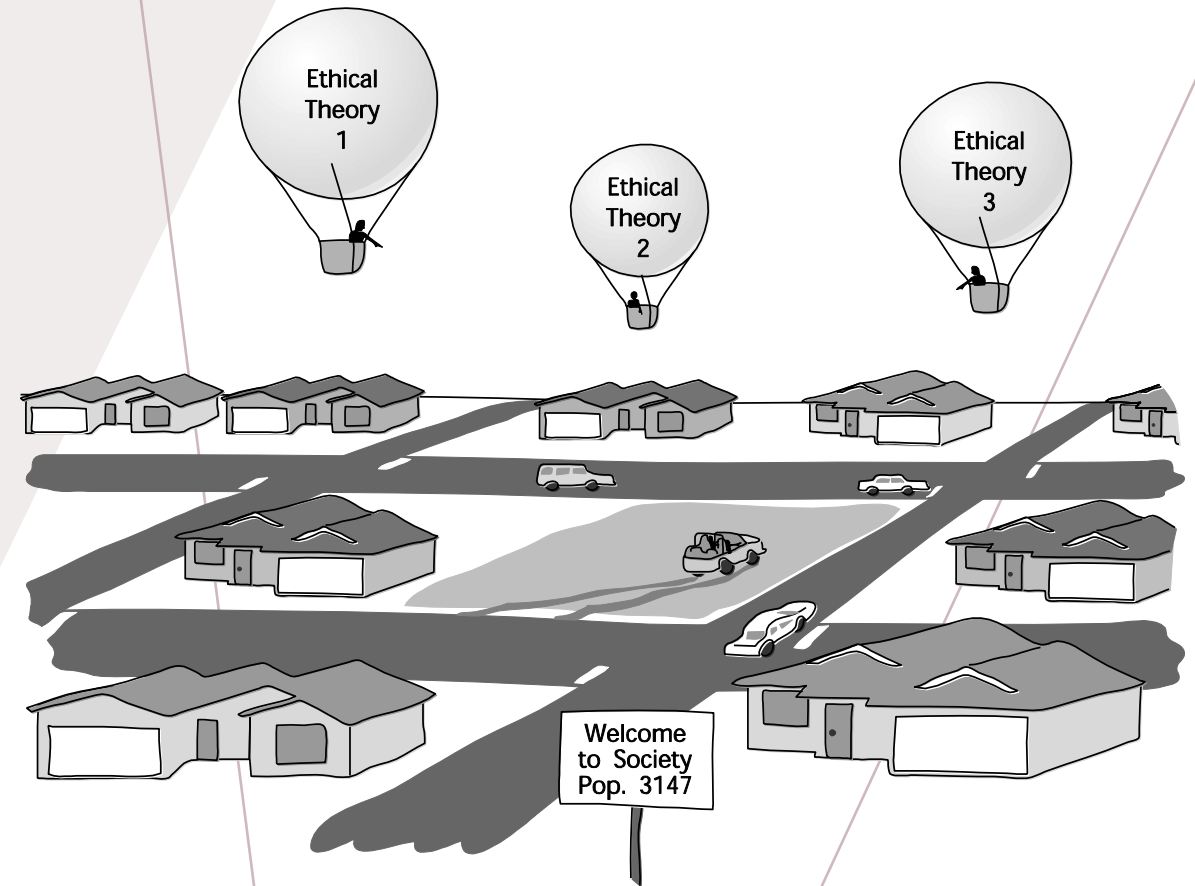
A society's/person's rules of conduct
What people ought / ought not to do in various situations



Ethics

Rational examination of morality
Evaluation of people's behavior

ANALOGY FOR DIFFERENCE BETWEEN MORALITY AND ETHICS



Source: Ethics for the Information Age by M. Quinn



WHY STUDY AI ETHICS?

- Numerous new issues accompany the rapid rise and ubiquitous use of AI in all walks of life, causing huge impacts
- Ethics: a way to help decide the “best” course of action
- “Common wisdom” may not be applicable
- To become responsible AI users and developers



SCENARIO 1

- Trying to deter speeding offenders, the Traffic Police installs AI-video cameras on all road junctions. The AI-video cameras can reliably detect cars traveling more than five km per hour above the speed limit, read license plate numbers and capture high-resolution pictures of vehicle drivers, in real time. Equipped with AI face recognition ability (99% accuracy), the system automatically issues a speeding ticket to the driver, complete with photo evidence. 3 months after the system is put into operation, the number of people speeding is reduced by 90 percent.
- The Organized Serious Crimes Bureau (OSCB) asks the Traffic Police for real-time access to the information collected by the AI-video cameras. The Traffic Police complies. 3 months later, OSCB uses this information to successfully arrest 5 wanted serious criminals.



SCENARIO 1

- Did the Traffic Police do anything wrong?
- Who benefited from the actions of the Traffic Police?
- Who was harmed by the actions of the Traffic Police?
- What other courses of action could the Traffic Police have taken to achieve its objectives?
- What additional information, if any, would help you answer the previous questions?

MORE ON ETHICS



Ethics: rational,
systematic analysis

"Doing ethics" means
explaining conclusions
Best explanations based
on facts, shared values,
logic



Ethics focuses on people's voluntary,
moral choices



Workable ethical theory produces
explanations that might be persuasive to
a skeptical, yet open-minded audience

*A GOOD ETHICAL
THEORY SUPPORTS
PERSUASIVE,
LOGICAL
ARGUMENTS*





SUBJECTIVE RELATIVISM

WHAT IS RELATIVISM?

Relativism

- No universal norms of right and wrong
- One person can say "X is right," another can say "X is wrong," and both can be right

Subjective relativism

- Each person decides right and wrong for himself or herself
- "What's right for you may not be right for me"

CULTURAL RELATIVISM

What is “right” and “wrong” depends upon a society’s actual moral guidelines

These guidelines vary from place to place and from time to time

A particular action may be right in one society at one time and wrong in other society or at another time

A variation of subjective realism

CASE FOR SUBJECTIVE RELATIVISM



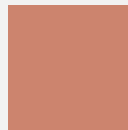
Well-meaning and intelligent people, and different (but equally successful) cultures may disagree on moral issues



Ethical debates are disagreeable and pointless



Different social contexts demand different moral guidelines



It is arrogant for one society to judge another



CASE AGAINST SUBJECTIVE RELATIVISM

- Blurs line between doing what you think is right and doing what you want to do
- Makes no moral distinction between the actions of different people
- Subjective realism and tolerance are two different things
- Decisions may not be based on reason
- Because two societies *do* have different moral views doesn't mean they *ought to* have different views
- It doesn't explain how moral guidelines are determined
- What if there are no cultural norms?
- It doesn't account for evolution of moral guidelines.
- Not a workable ethical theory



DIVINE COMMAND THEORY

OVERVIEW OF DIVINE COMMAND THEORY

Good actions: those aligned with God's will

Bad actions: those contrary to God's will

Holy books reveal God's will

We should use holy books as moral decision-making guides

DIVINE COMMAND THEORY IN ACTION



CASE FOR DIVINE COMMAND THEORY

We owe obedience to our Creator

God is all-good and all-knowing

God is the ultimate authority

CASE AGAINST DIVINE COMMAND THEORY



Different holy books disagree on certain teachings



Society is multicultural, secular



Some modern moral problems not directly addressed in scripture



"The good" $\neq$ "God" (equivalence fallacy – Bertrand Russell)



Based on obedience, not reason



Not a workable ethical theory for our purposes

The background features several thin, light-colored lines intersecting to form various geometric shapes. A large, light-colored triangle is positioned on the right side of the slide.

ETHICAL EGOISM

DEFINITION OF ETHICAL EGOISM

Each person should focus exclusively on his or her self-interest (aka Capitalism)

Morally right action: that action that provides self with maximum long-term benefit

Ayn Rand, author of *The Fountainhead* and *Atlas Shrugged*, espoused a theory akin to ethical egoism

CASE FOR ETHICAL EGOISM

It is practical since we are already inclined to do what's best for ourselves

It is better to let other people take care of themselves

The community can benefit when individuals put their well-being first

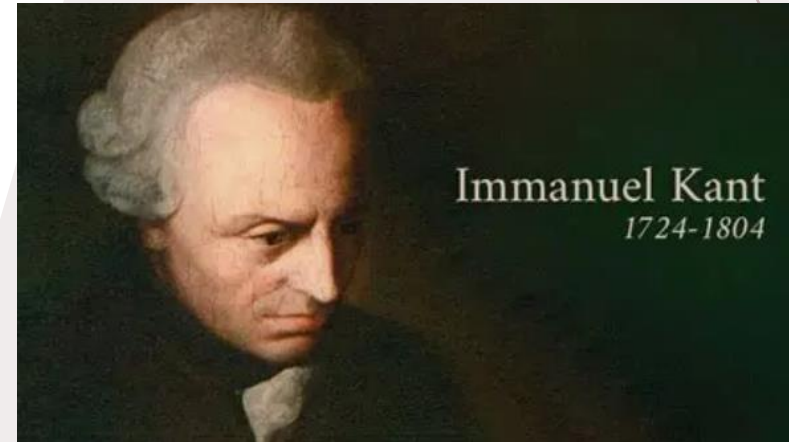
Other moral principles are rooted in the principle of self-interest



CASE AGAINST ETHICAL EGOISM

- An easy moral philosophy may not be the best moral philosophy
- We know a lot about what is good for someone else
- Self-interest can lead to blatantly immoral behavior
- Other moral principles are superior to principle of self-interest
- People who take the good of others into account lead happier lives
- By definition, it does not respect the “ethical point of view”
- Not a workable ethical theory

KANTIANISM *(DEONTOLOGY)*



CRITICAL IMPORTANCE OF GOOD WILL



Good will: the desire to do the right thing



Immanuel Kant: Only thing in the world that is good without qualification is a good will, a desire to act right for its own sake (i.e. intrinsic unconditional goodness, independent of external circumstances, consequences, judgement)



Reason and moral law (categorical imperative) should cultivate desire to do right thing

CATEGORICAL IMPERATIVE (1ST FORMULATION)

Act only from moral rules that you can, at the same time, will to be universal moral laws.

ILLUSTRATION OF 1ST FORMULATION

- Question: Can a person in dire straits make a promise with the intention of breaking it later?
- Proposed rule: "I may make promises with the intention of later breaking them."
- The person in trouble wants his promise to be believed so he can get what he needs.
- Universalize rule: Everyone may make & break promises
- Everyone breaking promises would make promises unbelievable, contradicting desire to have promise believed
- The rule is flawed. The answer is "No."

A QUICK CHECK



When evaluating a proposed action,
reverse roles



What would you think if that person
did the same thing to you?



Negative reaction → evidence that
your will to do that action violates the
Categorical Imperative

CATEGORICAL IMPERATIVE (2ND FORMULATION)

Act so that you treat both yourself and other people as ends in themselves, and never only as a means to an end.
(Treat people as rational moral agents, respect their dignity and free will)

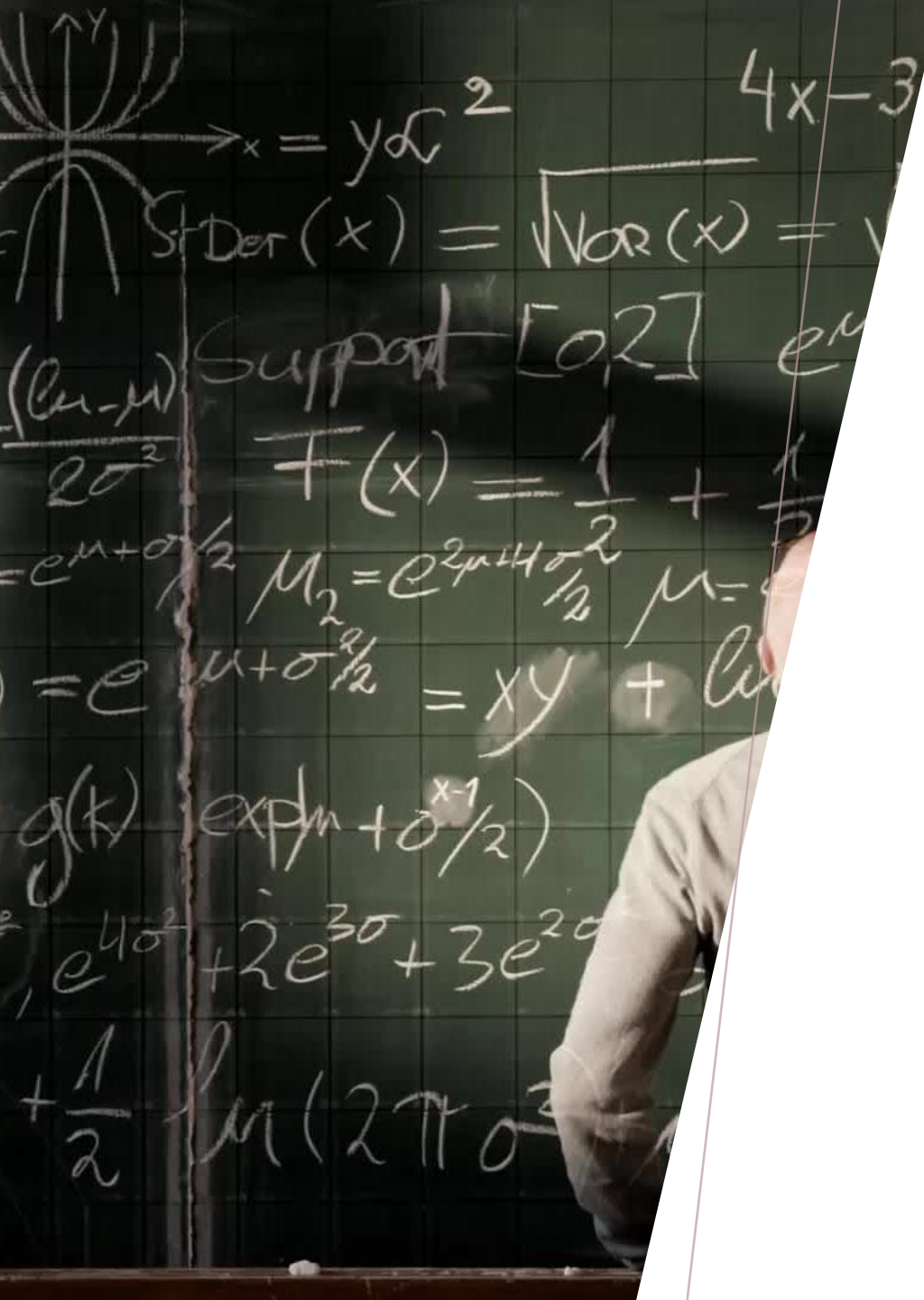


PLAGIARISM SCENARIO

- Mary
 - Single mother
 - Works full time
 - Takes two evening courses/semester
- AI class
 - Requires more work than normal
 - Carla earning an “A” on all work so far
 - Carla doesn’t have time to write final report
- Carla gets her boy friend (who has a PhD in AI) to write the report; submits it as her own work

KANTIAN EVALUATION (1ST FORMULATION)

- Mary wants credit for plagiarized report
- Rule: “You may claim credit for work performed by someone else”
- If rule universalized, reports would no longer be a credible indicator of student’s knowledge, and professors would not give credit for reports
- Proposal moral rule is self-defeating
- Mary is morally wrong



KANTIAN EVALUATION (2ND FORMULATION)

- Mary submitted another person's work as her own
- She attempted to deceive professor
- She treated professor as a means to an end
 - End: passing the course
 - Means: manipulate professor
- What Mary did was morally wrong

CASE FOR KANTIANISM

Aligns with common moral concern:
“What if everyone acted that way?”

Produces universal moral guidelines

Treats all persons as moral equals
Duty over emotion

PERFECT AND IMPERFECT DUTIES



Perfect duty: duty obliged to fulfill without exception

Example: Telling the truth



Imperfect duty: duty obliged to fulfill in general but not in every instance

Example: Helping others

These are **strict moral obligations** that must always be followed. Violating them leads to logical contradiction if universalized.

- **To oneself:**
 - Do not commit suicide.
 - Develop one's talents.
- **To others:**
 - Do not lie.
 - Do not steal.
 - Do not break promises.

Example: You must never lie, even to protect someone, because lying cannot be universalized without undermining trust.

EXAMPLES OF PERFECT DUTIES

These are **flexible duties** that allow for discretion in how and when they are fulfilled, but they are still morally required.

- **To oneself:**
 - Cultivate moral and intellectual virtues.
- **To others:**
 - Help others in need.
 - Promote the happiness of others.

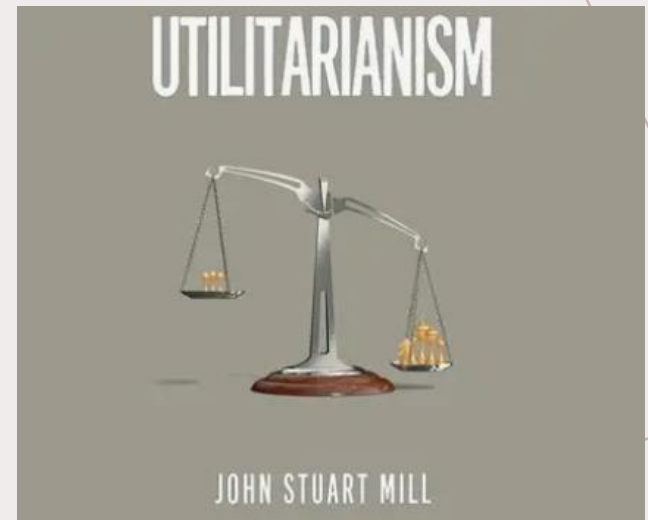
Example: You are morally obligated to help others, but you can choose how and when to do so.

INPERFECT DUTIES

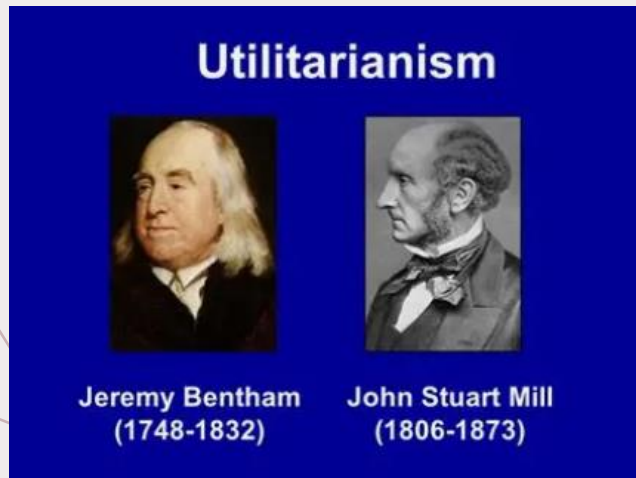
CASE AGAINST KANTIANISM

- Sometimes no rule adequately characterizes an action (moral dilemma)
- Sometimes there is no way to resolve a conflict between rules
 - In a conflict between a perfect duty and an imperfect duty, perfect duty prevails
 - In a conflict between two perfect duties, no solution
- Kantianism allows no exceptions to perfect duties
- Rules need to be interpreted (e.g. do not kill)
- Despite weaknesses, a workable ethical theory

ACT UTILITARIANISM



PRINCIPLE OF UTILITY



- Jeremy Bentham and John Stuart Mill
- An action is good if its benefits exceeds its harms
- An action is bad if its harms exceed its benefits
- Utility: tendency of an object to produce happiness or prevent unhappiness for an individual or a community
- Happiness = advantage = benefit = good = pleasure = well-being
- Unhappiness = disadvantage = cost = evil = pain

PRINCIPLE OF UTILITY (GREATEST HAPPINESS PRINCIPLE)



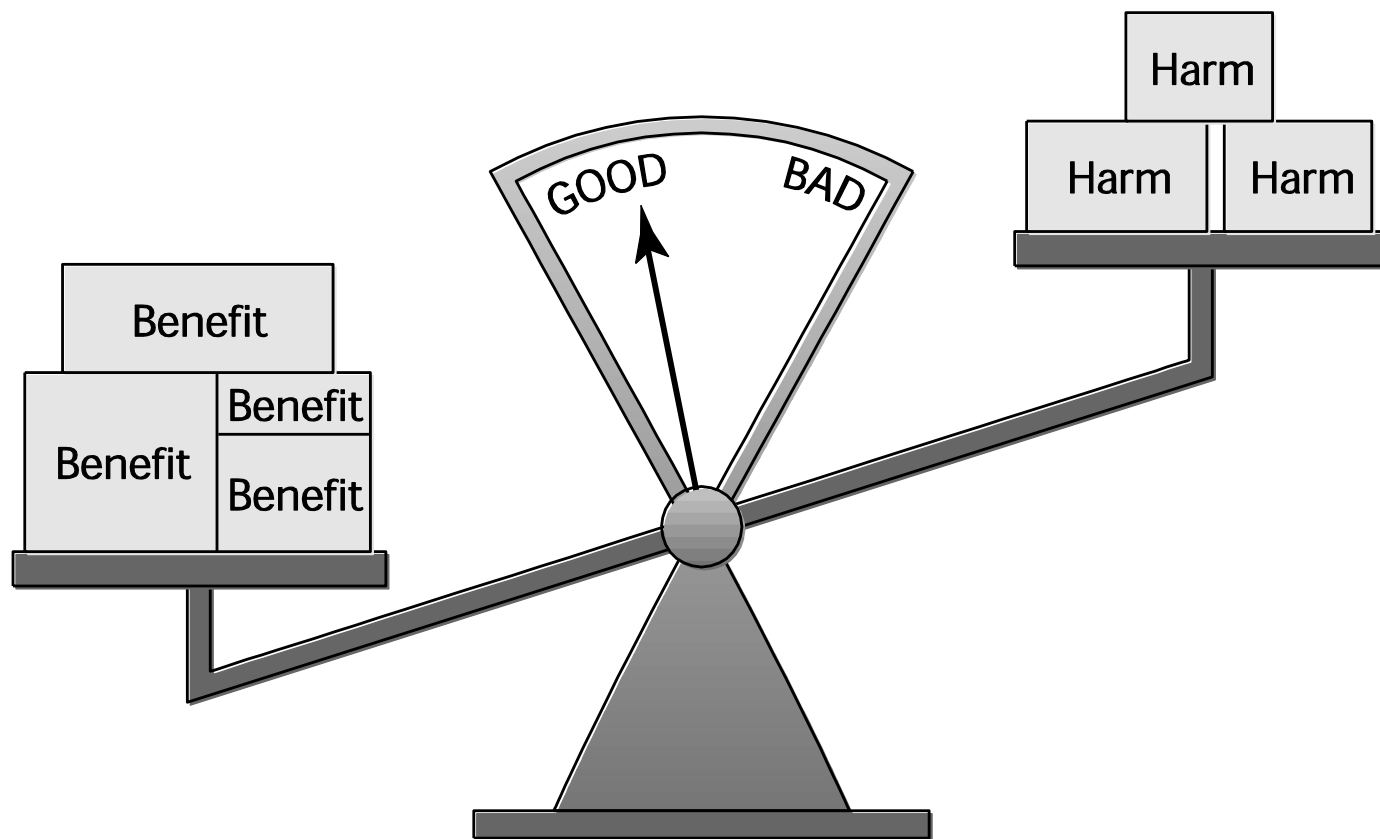
An action is right (or wrong) to the extent



that it increases (or decreases) the



total happiness of the affected parties.



PRINCIPLE OF UTILITY

ACT UTILITARIANISM

- Utilitarianism
 - Morality of an action has nothing to do with intent
 - Focuses on the consequences
 - A consequentialist theory
- Act utilitarianism
 - Add up change in happiness of all affected beings
 - $\text{Sum} > 0$, action is good
 - $\text{Sum} < 0$, action is bad
 - Right action to take: one that maximizes the sum

*BENTHAM:
WEIGHING
PLEASURE/PAIN
(HEDONIC
CALCULUS)*

HOW TO MEASURE PLEASURE?

- **Intensity:** How strong is the pleasure?
- **Duration:** How long will the pleasure last?
- **Certainty** or uncertainty: How likely or unlikely is it that the pleasure will occur?
- **Propinquity** or remoteness: How soon will the pleasure occur?
- **Fecundity:** Potential to generate more pleasure.
- **Purity:** The probability that it will not be followed by sensations of the opposite kind.
- **Extent:** How many people will be affected?

HIGHWAY ROUTING SCENARIO

- State may replace a curvy stretch of highway
- New highway segment 2 km shorter
- 150 houses would have to be removed
- Some wildlife habitat would be destroyed

EVALUATION

- Costs
 - \$20 million to compensate homeowners
 - \$10 million to construct new highway
 - Lost wildlife habitat worth \$1 million
- Benefits
 - \$39 million savings in automobile driving costs
- Conclusion
 - Benefits exceed costs
 - Building highway is a good action

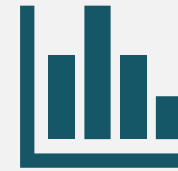
CASE FOR ACT UTILITARIANISM



FOCUSES ON HAPPINESS



DOWN-TO-EARTH
(PRACTICAL)



COMPREHENSIVE

CASE AGAINST ACT UTILITARIANISM

- Unclear whom to include in calculations and how far out into the future to consider
- Difficulty in measurement
- Too much work (every action needs to be individually evaluated)
- Ignores our innate sense of duty or moral urgency
- We cannot predict consequences with certainty
- Susceptible to the problem of moral luck (i.e. being assigned moral praise/blame for consequences out of one's control)
- Overall, a workable ethical theory

*RULE
UTILITARIANISM -
APPLYING
PRINCIPLE OF
UTILITY TO RULES*

We ought to adopt moral rules which, if followed by everyone, will lead to the greatest increase in total happiness

Act utilitarianism applies Principle of Utility to individual actions

Rule utilitarianism applies Principle of Utility to moral rules

Bentham and Mill

Compare / Contrast / Compare / Contrast / Compare / Contrast

BENTHAM



J.S. MILL

Principle of Utility



Greatest Happiness Principle

Emphasis on **pleasure**



Emphasis on **happiness**

Concerned with **QUANTITY** of **pleasure** "*Push-pin [a simple child's game] is as good as poetry*"



Concerned with **QUALITY** of **pleasure** "...better to be a Socrates dissatisfied than a fool satisfied"

Social Reformer: criminal, judicial, penal



Social Reformer: equality for women

Labelled an **act utilitarian**



Labelled a **rule utilitarian**

Feature	Act Utilitarianism	Rule Utilitarianism
Core Principle	Evaluate each action by its direct consequences	Follow rules that generally lead to the greatest good
Decision Basis	Case-by-case analysis	General rules derived from utility
Flexibility	Highly flexible; adapts to each situation	Less flexible; adheres to established rules
Moral Guidance	No fixed rules; depends on context	Rules provide consistent moral guidance
Risk	Can justify morally questionable acts if they maximize utility	Avoids justifying harmful acts by relying on stable rules
Example in Dilemma	Save the person whose survival leads to greatest happiness in that moment	Follow a rule like "protect your spouse first" if it generally promotes well-being

RULE VS ACT UTILITARIANISM

CASE FOR RULE UTILITARIANISM

- Not every moral decision requires performing utilitarian calculus
- Moral rules survive exceptional situations
- Avoids the problem of moral luck
- Reduces the problem of bias
- Appeals to a wide cross-section of society

CASE AGAINST UTILITARIANISM IN GENERAL

- All consequences must be measured on a single scale.
 - All units must be the same in order to do the sum
 - In certain circumstances utilitarians must quantify the value of a human life
- Utilitarianism ignores the problem of an unjust distribution of good consequences.
 - Utilitarianism does **not** mean “the greatest good of the greatest number”
 - That requires a principle of justice
 - What happens when a conflict arises between the Principle of Utility and a principle of justice?
- Despite weaknesses, both act utilitarianism and rule utilitarianism are workable ethical theories



SOCIAL CONTRACT THEORY

BASIS OF SOCIAL CONTRACT THEORY

- Thomas Hobbes
 - In a “state of nature,” our lives would be “solitary, poor, nasty, brutish, and short”
 - We *implicitly* accept a social contract
 - Establishment of moral rules to govern relations among citizens
 - Government capable of enforcing these rules
- Jean-Jacques Rousseau
 - In an ideal society, no one above the rules
 - That prevents society from enacting bad rules

*JAMES
RACHELS'S
DEFINITION*

“Morality consists in the set of rules,

governing how people are to

treat one another, that rational

people will agree to accept, for their

mutual benefit, on the condition that

others follow those rules as well.”

KINDS OF RIGHTS

- Negative right: A right that another can guarantee by leaving you alone
- Positive right: A right obligating others to do something on your behalf
- Absolute right: A right guaranteed without exception
- Limited right: A right that may be restricted based on the circumstances

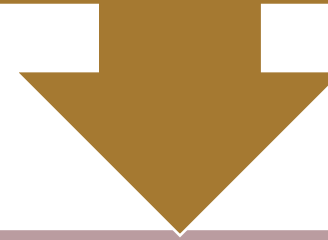
*CORRELATION
BETWEEN
TYPES OF
RIGHTS*

Positive rights tend
to be more limited

Negative rights tends
to be more absolute

JOHN RAWLS'S PRINCIPLES OF JUSTICE

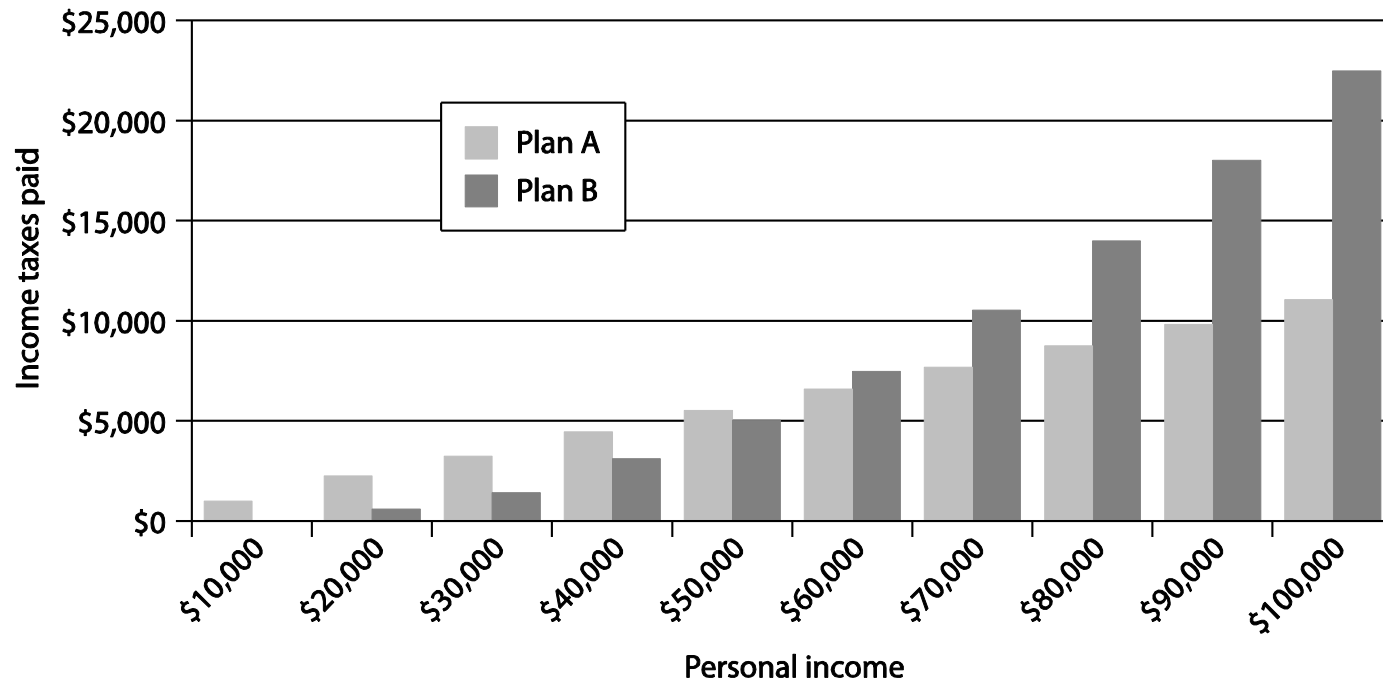
Each person may claim a “fully adequate” number of basic rights and liberties, so long as these claims are consistent with everyone else having a claim to the same rights and liberties



Any social and economic inequalities must

Be associated with positions that everyone has a fair and equal opportunity to achieve

Be to the greatest benefit of the least-advantaged members of society (the difference principle)



RAWLS'S DIFFERENCE PRINCIPLE

CASE FOR SOCIAL CONTRACT THEORY

- Framed in language of rights
- Explains why people act in self-interest in the absence of common agreement
- Provides clear analysis of certain citizen/government problems
 - Why okay for government to deprive criminals of certain rights
 - How civil obedience could be morally justified
- Workable ethical theory

*CASE
AGAINST
SOCIAL
CONTRACT
THEORY*

No one signed social contract

Some actions have multiple characterizations

Conflicting rights problem

May unjustly treat people incapable of upholding contract

Despite weaknesses, a workable theory

VIRTUE ETHICS



A blurred background image showing several people in a room, with their hands raised and clapping, suggesting an audience or a group of people in a meeting or presentation.

CRITIQUE OF ENLIGHTENMENT THEORIES

- Kantianism, utilitarianism, social contract theory ignore important moral considerations
 - moral education
 - moral wisdom
 - family and social relationships
 - role of emotions
- Virtue ethics
 - *arete*, virtue, excellence: reaching highest potential
 - Aristotle's *Nicomachean Ethics* (4th century BC)

VIRTUES AND VICES

Two types of virtue

- intellectual virtues: virtues associated with reasoning and truth
- moral virtues: virtues of character (e.g., honesty)

Moral virtues

- developed by habitually performing right action
- deep-seated character traits
- disposition to act in a certain way *and* feel in a certain way

*ARISTOTLE:
HAPPINESS DERIVES
FROM LIVING A LIFE
OF VIRTUE.*



VICES

- Vices are opposite of virtues
- Vice: a character trait that prevents a human being from flourishing or being truly happy
- Often, a virtue situated between two vices
 - Courage between cowardliness and rashness
 - Generosity between stinginess and prodigality



SUMMARY OF VIRTUE ETHICS

- A right action is an action that a virtuous person, acting in character, would do in the same circumstances.
- A virtuous person is a person who possesses and lives out the virtues.
- The virtues are those character traits human beings need in order to flourish and be truly happy.

Virtue	Description
Courage	Facing fears and taking appropriate risks
Temperance	Practicing self-control and moderation
Justice	Treating others fairly and giving each their due
Prudence (Practical Wisdom)	Making wise decisions based on reason and experience
Honesty	Being truthful and transparent
Compassion	Showing empathy and concern for others
Generosity	Willingness to give and share with others
Loyalty	Being faithful to commitments and relationships
Humility	Recognizing one's limitations and avoiding arrogance
Perseverance	Persisting through challenges and adversity

CORE VIRTUES (ARISTOTLE)

Virtue (Chinese)	Translation	Meaning
仁 (Rén)	Benevolence / Humaneness	Compassion, empathy, and kindness toward others
义 (Yì)	Righteousness / Justice	Doing what is morally right, beyond self-interest
礼 (Lǐ)	Ritual / Propriety	Respect for tradition, manners, and social harmony
智 (Zhì)	Wisdom	Moral discernment and understanding of right and wrong
信 (Xìn)	Integrity / Trustworthiness	Being honest and keeping one's word

CONFUCIOUS
VIRTUES
賢德

CASE FOR VIRTUE ETHICS

- It often makes more sense to focus on virtues than obligations, rights, or consequences
- Personal relationships can be morally relevant to decision making
- Our moral decision-making skills develop over time
- With this theory there are no irresolvable moral dilemmas
- Emotions play an important role in living a moral life

CASE AGAINST VIRTUE ETHICS

- Reasonable people may disagree on character traits needed for human flourishing
- Cannot use virtue ethics to guide government policy
- Virtue ethics undermines attempts to hold people responsible for their bad actions
- Despite weaknesses, virtue ethics a workable theory

COMPARING WORKABLE ETHICAL THEORIES



OBJECTIVISM VS. RELATIVISM



Objectivism: Morality has an existence outside the human mind



Relativism: Morality is a human invention



Divine command theory, ethical egoism, Kantianism, utilitarianism, social contract theory, and virtue ethics examples of objectivism

WORKABLE ETHICAL THEORIES

We seek theories with these characteristics:

- Based on the ethical point of view
- Objective moral principles developed using logical reasoning based on facts and commonly held values

Workable ethical theories

- Kantianism
- Act and rule utilitarianism
- Social contract theory
- Virtue ethics

WORKABLE ETHICAL THEORIES

KANTIAN



Follow
universal
moral laws

UTILITARIAN



Maximize
overall
happiness

VIRTUE



Cultivate
moral
character

SOCIAL CONTRACT



Follow
mutually
agreed rules

Kantian Ethics

Focuses on duty and universal moral laws, emphasizing intentions behind actions and respect for individuals.

Utilitarianism

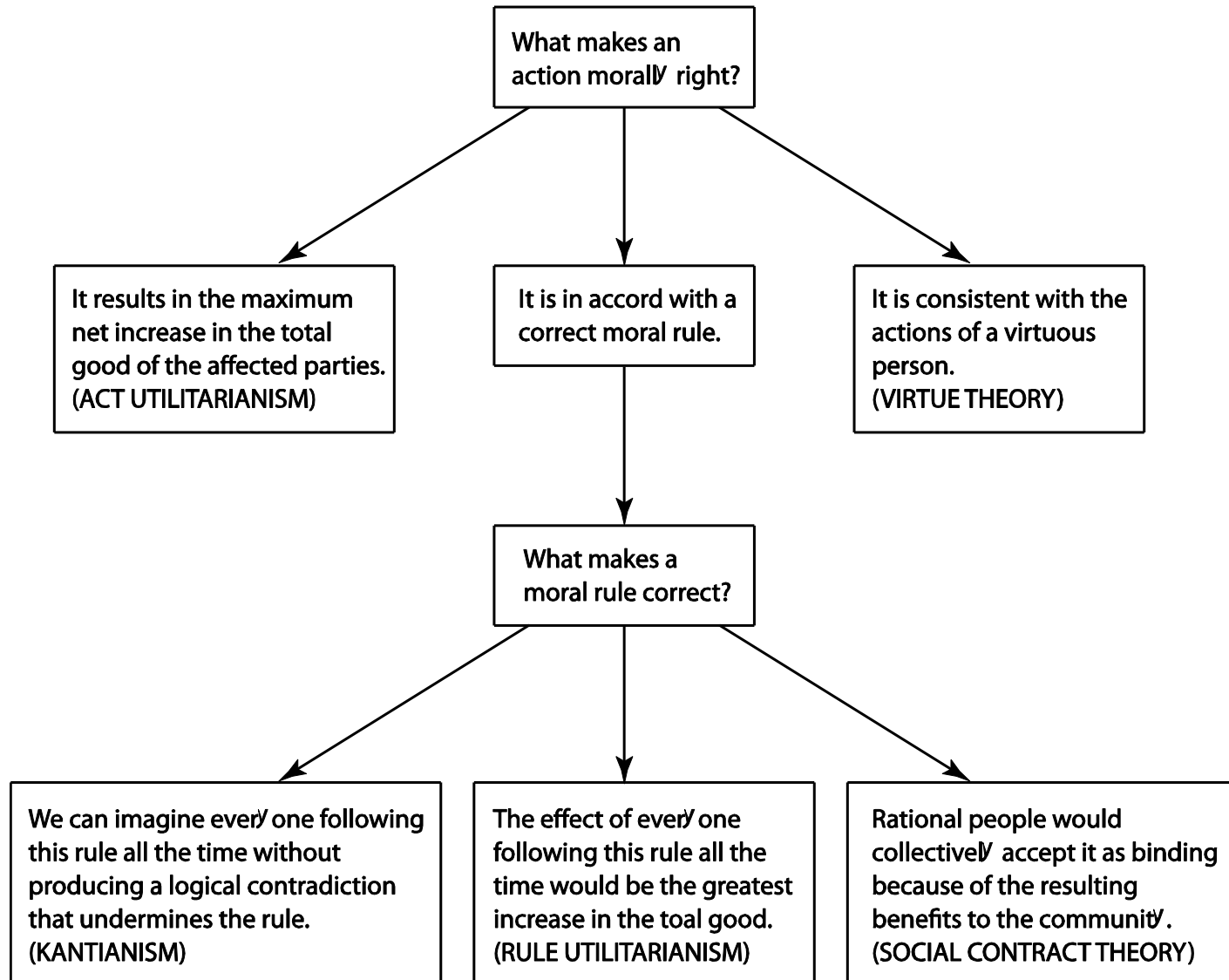
Aims for the greatest good for the greatest number, focusing on outcomes and promoting overall well-being.

Virtue Ethics

Centers on moral character and virtues like courage and honesty, promoting holistic moral development.

Social Contract Theory

Bases morality on mutual societal agreements, emphasizing justice, fairness, and cooperative rights.



COMPARING WORKABLE ETHICAL THEORIES

SUMMARY

INSIGHTS OFFERED BY VARIOUS THEORIES

Kantianism: Every person is equally valuable, and when you interact with other people you should always respect them as rational beings.

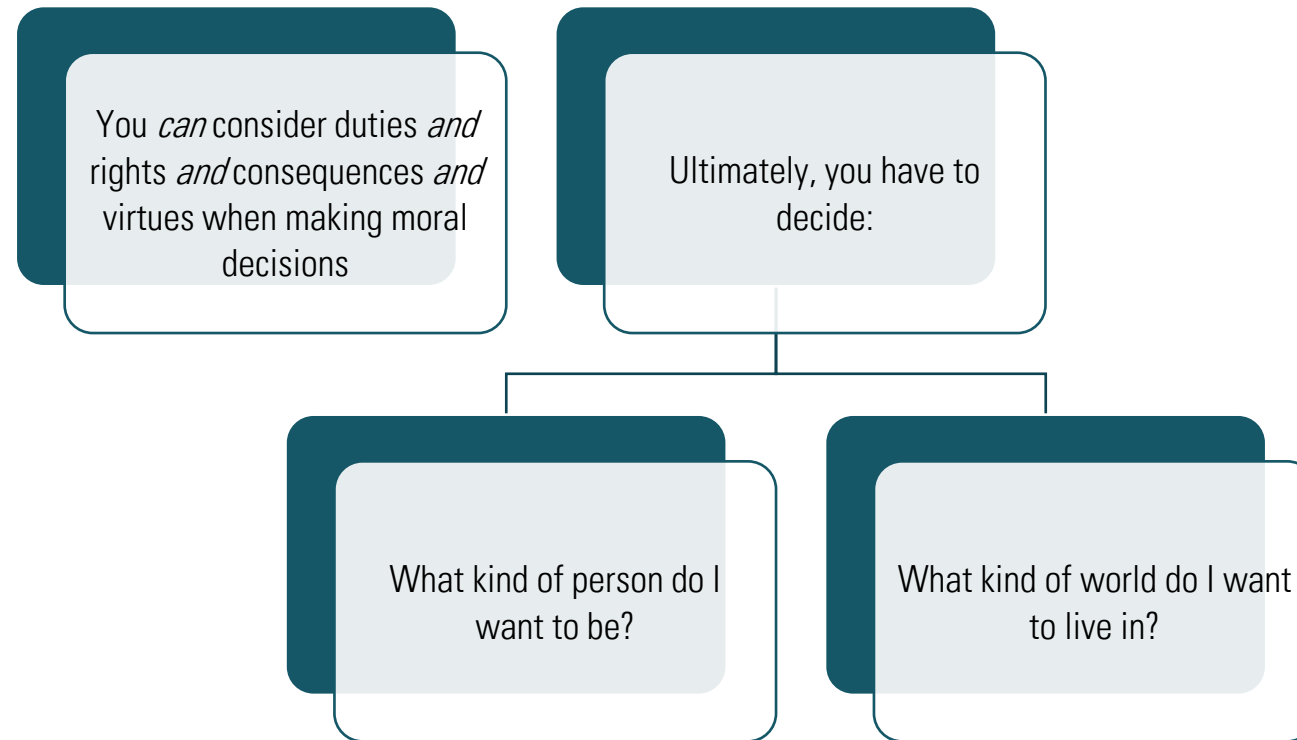
Utilitarianism: You should consider the consequences of an action before deciding whether it's right or wrong.

Social contract theory: We should collectively promote human rights, such as the rights to life, liberty, and property.

Virtue ethics: You can count on a good person to do the right thing at the right time in the right way.

Ethical Framework	Core Principle	Key Thinkers	Moral Focus	Strengths	Criticisms
<i>Kantian Ethics</i>	Duty and universal moral laws	Immanuel Kant	Intentions and adherence to duty	Emphasizes consistency, respect for persons, and moral autonomy	Can be rigid; ignores consequences
<i>Utilitarianism</i>	Greatest good for the greatest number	Jeremy Bentham, John Stuart Mill	Outcomes and consequences	Practical, outcome-focused, promotes overall well-being	Can justify harmful actions if they benefit the majority
<i>Virtue Ethics</i>	Cultivation of moral character	Aristotle	Character and virtues (e.g., courage, honesty)	Holistic, focuses on moral development and personal growth	Lacks clear guidance for specific actions
<i>Social Contract Theory</i>	Morality based on mutual agreement in society	Thomas Hobbes, John Locke, Jean-Jacques Rousseau	Justice, rights, and societal rules	Grounds ethics in fairness and cooperation	Assumes rational agreement; may exclude marginalized voices

ETHICAL EXPLANATIONS THAT MIGHT BE PERSUASIVE TO A SKEPTICAL, YET OPEN-MINDED AUDIENCE





ASSIGNMENT (DUE AT NOON, 22 SEPT)

- Trying to deter speeding offenders, the Traffic Police installs AI-video cameras on all road junctions. The AI-video cameras can reliably detect cars traveling more than five km per hour above the speed limit, read license plate numbers and capture high-resolution pictures of vehicle drivers, in real time. Equipped with AI face recognition ability (99% accuracy), the system automatically issues a speeding ticket to the driver, complete with photo evidence. 3 months after the system is put into operation, the number of people speeding is reduced by 90 percent.
- The Organized Serious Crimes Bureau (OSCB) asks the Traffic Police for real-time access to the information collected by the AI-video cameras. The Traffic Police complies. 3 months later, OSCB uses this information to successfully arrest 5 wanted serious criminals.



ANSWER IN 1-2 PAGES

- Did the Traffic Police do anything wrong?
- Who benefited from the actions of the Traffic Police?
- Who was harmed by the actions of the Traffic Police?
- What other courses of action could the Traffic Police have taken to achieve its objectives?
- What additional information, if any, would help you answer the previous questions?